

Practical 1

Aim: Set up Python using a virtual environment, create basic variables, and demonstrate different data types (int, float, string, list, tuple, dictionary, set).

Source-Code:

```
# 1. Integer (int)
age = 22
print("Age:", age)
print("Type of age:", type(age))

# 2. Float
height = 5.11
print("Height:", height)
print("Type of height:", type(height))

print("-----")

# 3. String
name = "NotGonnaTell"
print("Name:", name)
print("Type of name:", type(name))

print("-----")

# 4. List (Ordered, Changeable)
fruits = ["apple", "banana", "mango"]
print("Fruits List:", fruits)
print("Type of fruits:", type(fruits))

print("-----")

# 5. Tuple (Ordered, Not Changeable)
coordinates = (10, 20)
print("Coordinates Tuple:", coordinates)
print("Type of coordinates:", type(coordinates))

print("-----")

# 6. Dictionary (Key-Value Pair)
student = {
    "name": "NotGonnaTell",
    "age": 22,
    "course": "MCA"
}
print("Student Dictionary:", student)
print("Type of student:", type(student))

print("-----")

# 7. Set (Unordered, Unique values)
numbers = {1, 2, 3, 3, 4}
print("Set of numbers:", numbers)
print("Type of numbers:", type(numbers))
```

Output:

Age: 22

Type of age: <class 'int'>

Height: 5.11

Type of height: <class 'float'>

Name: NotGonnaTell

Type of name: <class 'str'>

Fruits List: ['apple', 'banana', 'mango']

Type of fruits: <class 'list'>

Coordinates Tuple: (10, 20)

Type of coordinates: <class 'tuple'>

Student Dictionary: {'name': 'NotGonnaTell', 'age': 22, 'course': 'MCA'}

Type of student: <class 'dict'>

Set of numbers: {1, 2, 3, 4}

Type of numbers: <class 'set'>